

Wrocław University of Science and Technology

Faculty of Information and Communication Technology

Field of study: **Computer Engineering (ITE)**

Speciality: **Computer Graphics and Multimedia Systems (IGM)**

MASTER THESIS

**Analysis of deep neural network classification errors in
image recognition, taking into account semantic similarity**

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Keywords:

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Summary

Place a concise summary of your thesis here. It should include a description of the purpose of the work, the methods used, and the results obtained.

Both summaries should fit on a single page.

Abstract

Place a concise summary of your thesis here. It should include a description of the purpose of the work, the methods used and the results obtained.

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List of Acronyms

DRY Don't repeat yourself	7
KISS Keep It Simple, Stupid	7

Chapter 1

Introduction

Write a short introduction to your thesis here. You can discuss the basic algorithms related to the thesis objective. You can also provide a short historical background. Remember to keep the text concise and avoid generic statements that add no value. Here is an example opening:

The problem of specialized algorithms for stream data analysis attracted significant researcher interest at the beginning of the twentieth century. One of the first such algorithms was used to select a random element according to a uniform distribution from a window consisting of the last N stream elements, where N is a fixed number. ...

1.1. Goal and Thesis Structure

Try to describe the thesis objective here as concisely and precisely as possible. Here is an example beginning of this section:

In this thesis, we study the problem of determining the k most frequent elements in a window containing the last N elements of a data stream. ...

Thesis Structure

Briefly describe, in order, all chapters of your thesis. Example:

In the first chapter (see 1.3), we begin with a description of requirements imposed on streaming algorithms (the main requirement is logarithmic memory usage with respect to stream size), describe the concept of a sliding window in data streams, and discuss the Braverman algorithm,

In the second chapter, we discuss ...

...

Technologies Used

Describe which programming languages and compilers you used, in which environment you ran and tested the developed software, and identify the most important libraries used in the implemented programs.

1.2. Use of AI

If modern AI tools were used during the thesis work, this section should describe them: which AI tools were used, what they were used for, and which parts of the thesis they affected. **You**

must not omit this section, although you may place it elsewhere in the thesis. Be sure to read the “Guidelines on AI Usage Rules”, available on the Faculty website (see [\[1\]](#)).

If AI was used to generate code, mention that here as well, and mark those fragments in the program code (in comments) where AI intervened. Example:

The thesis used Google’s Gemini 3.0 assistant in the Pro version to help transform code from C into Haskell (see, among others, listing 2.2) and to assist in documenting all modules developed during the thesis.

If AI was not used during the thesis work, include a sentence such as: "*AI was not used in this thesis.*"

1.3. Background

Here you can describe the basic concepts, terms, and notation used in the thesis.

1.4. Theoretical Foundations

You may provide the main results used in your thesis. These can include references to the literature - chapter 2.4 explains bibliography formatting and citation methods used in the thesis (`\cite{...}`, `\citeauthor{...}`, ...)

Chapter 2

About the Template

This chapter contains several tips that may help while writing your thesis, as well as examples of using LaTeX packages included in this template. The template is based on the classic report class. The most important component of this template is the `settings.tex` file. Here are some details:

1. All thesis files should be saved in UTF-8 format.
2. We use fonts from the `newtxtext` and `newtxmath` families. These are modern Times-style fonts. The document's main font is serif. The title page is displayed in a sans-serif variant.
3. We use the `biblatex` package to display bibliography, and the `biber` program to process bibliography data. This is a modern replacement for the older `bibtex` program. Before using this template, learn how to configure your LaTeX build environment so that it uses `biber`. If you have problems, try running the command `biber file` from the command line, where `file` is the main project file name without extension. If you use [Overleaf](#), it should detect `biber` usage automatically.
4. If you use local LaTeX environments (e.g., TeXnicCenter), the first compilation with this template may take some time because the environment may install required packages.
5. The template imports the `listings` package for displaying program listings and the `algorithm` and `algpseudocode` packages for displaying pseudocode.
6. The template imports the `acro` package for handling synonyms/acronyms. If you do not use synonyms, comment out two lines in the main file (they are described in comments on the main template page). Put your synonyms in a dedicated glossary file.
7. The template uses the `cleveref` package for more flexible in-document references. Check this package's capabilities: for example, compare using `\cref{..}` to the standard `\ref{..}` when referencing figures or tables.

Split your thesis into a series of smaller files (for example: `Chapter01.tex`, `Chapter02.tex`, etc.). Include these chapters in the main thesis file using **`include`** or **`input`** commands (`include` forces a page break). Place chapters in the `chapters` subdirectory.

2.1. User Definitions

If you use the same text fragment several times, it is worth defining your own LaTeX commands. Here is an example of using definitions such as `\newcommand{\NN}{\mathbb{N}}` from the main template file. Example:

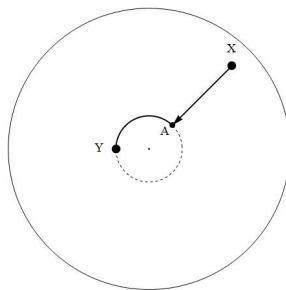
We use the symbol $\mathbb{R}$ to denote the set of real numbers and the symbol $\mathbb{N}$ to denote the set of natural numbers. Note that we include the number 0 in the set of natural numbers, i.e., $\mathbb{N} = \{0, 1, 2, \dots\}$.

Using such definitions makes thesis text more consistent and eliminates many errors.

2.2. Illustrations

The `\includegraphics` command allows you to insert images into a LaTeX document (this requires the `graphicx` package included in the template). Place images in the `figures` subdirectory.

You can find a detailed description of this command's parameters at https://latexref.xyz/_005cincludegraphics.html.



A more flexible approach is to wrap graphics in a `picture` construction (see Fig. Fig. 2.1). Figure Fig. 2.1 appears below.

This thesis template uses the `cleveref` package for flexible references within the document. For example, when using the `cref` command (instead of standard `\ref{fig:Lines}`), you do not need to type the `Fig.` prefix manually.

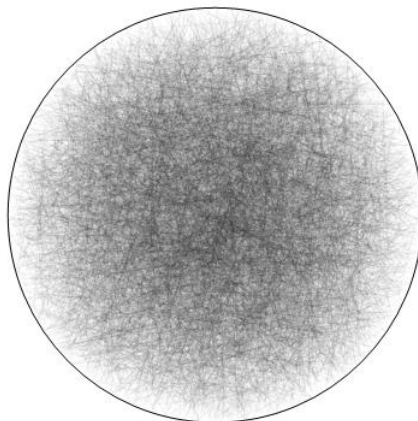


Figure 2.1. Simulation of 10000 random line segments in the unit circle. The figure was generated independently.

If you uncomment the `\listoffigures` command in the main file, this figure will appear in the generated list of figures.

2.3. Mathematical Formulas

Inline mathematical expressions are formatted with `$ ... $`, e.g. $f(x) = \sin(x^2)$. With `\[ ... \]` you format centered, unnumbered equations, e.g.

$$\sin^2(\alpha) + \cos^2(\alpha) = 1 ,$$

and with `\begin{equation} ... \end{equation}` you format numbered equations, e.g.

$$\exp(I \cdot t) = \cos(t) + I \cdot \sin(t) \tag{2.1}$$

Use numbered equations **only when** you will reference them later in the text using `\ref{...}`.

If your thesis includes any theorems or lemmas, use the `theorem`, `lemma`, `corollary`, and `proof` environments defined in the `amsthm` package imported by this template. Example:

Theorem 2.1 (Euclid). *There are infinitely many prime numbers.*

Proof. Assume that $p_1, \dots, p_n$ is the list of all prime numbers. Let

$$q = (p_1 \cdot \dots \cdot p_n) + 1 .$$

Then $q > 1$, so there must exist a prime number p such that $p|q$

□

2.4. Citations

In this template we use the `biber` library for citation formatting. It is a more modern version of the standard (but no longer actively developed) `bibtex` library. Example usage:

The book [2] is devoted to fundamental algorithms. Author Knuth discusses most classical algorithms in it. From this thesis perspective, the algorithm in Knuth [3, p. 249] is important.

Papers [4], [5] contain concise introductions to *geographic routing*.

Entries describing bibliographic sources are best kept in a separate `bibtex` file. The template includes an example `bibliography.bib` file containing examples of basic entry types. Well-formatted bibliographic entries can be found, for example, on [dblp](#) and in [Google Scholar](#).

Biber, used with `biblatex`, provides many predefined styles for citation and bibliography formatting, with common styles such as `numeric`, `alphabetic`, `authoryear`, `authortitle`, and `verbatim`, along with sorting options such as `nty` (name, title, year) or `nyt` (name, year, title). These styles are selected with the `style=` option in

```
\usepackage[backend=biber, style=...]{biblatex}
```

in the LaTeX preamble. Choose the formatting style jointly with your thesis supervisor. In `main.tex`, a numeric style without sorting has been proposed, which means the bibliography is displayed in the order in which items are cited in the thesis text.

Note

In LaTeX, you can manage citations manually without a bibtex file. However, manually formatting bibliography correctly is quite tedious. Also remember that every item appearing in the reference list must be cited in the thesis text.

2.5. Tables

You insert tables in the standard way. After uncommenting `\listoftables` in the main template file, a list of tables will be included in the document. You can refer to this table using `\cref{tab:example}` (Tab. 2.1).

Table 2.1. Sample table

Category	Parameter	Result
Test A	Low load	15 ms
Test B	High load	120 ms

2.6. Procedure Code

The `listings` package is included in this template. It makes it easier to insert source code into the document. In this template, in `settings.tex`, default display parameters for code are configured.

Here is an example usage:

```
1  /*
2   * Euclidean algorithm.
3   * Returns gcd(a, b).
4   */
5  int gcd(int a, int b)
6  {
7      while (b != 0) {
8          int r = a % b;
9          a = b;
10         b = r;
11     }
12     return a;
13 }
```

Listing 2.1. Euclidean algorithm in C

Here is another example, a Haskell QuickSort implementation (listing 2.2) that uses the *list comprehension* mechanism to improve code readability:

```
1  -- QuickSort: time O(n^2); average time O(n log(n))
2  qs [] = []
3  qs (x:xs) = qs [t | t <- xs, t < x] ++
```


References

- [1] PWr WIT. *AI usage guidelines at WIT*. 2025. URL: https://wit.pwr.edu.pl/wydzial/jakosc_ksztalcenia/wytyczne-dot-narzedzia-si-na-wit.
- [2] Donald E. Knuth. *The art of computer programming. Volume 1: Fundamental Algorithms*. 3rd. Vol. 1. The art of computer programming. Reading, Massachusetts: Addison-Wesley, 1997.
- [3] Donald E. Knuth. *The art of computer programming. Volume 2: Seminumerical Algorithms*. 3rd. Vol. 2. The art of computer programming. Reading, Massachusetts: Addison-Wesley, 1997.
- [4] Fabian Kuhn, Roger Wattenhofer, and Aaron Zollinger. „An algorithmic approach to geographic routing in ad hoc and sensor networks”. In: *IEEE/ACM Trans. Netw.* 16.1 (2008), pp. 51–62.
- [5] L. Popa et al. „Balancing Traffic Load in Wireless Networks with Curveball Routing”. In: *Mobihoc’07*. 2007.

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